

Sean Bangalore

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SUMMARY

Engineering leader for high-throughput, mission-critical distributed platforms. Technical owner of Apple's core ML orchestration layer, serving millions of requests per day; cut infrastructure costs by more than \$10M/year through multi-region caching and serving-stack optimization. Lead principal-level engineers and multi-team initiatives across roadmap, architecture, SLO/SLI standards, incident response, and self-service platforms. Specialized in large-scale LLM inference. Java, Kubernetes, AWS, multi-region architecture.

EXPERIENCE

Senior Machine Learning Engineer — Tech Lead, ML Platform, Apple Aug 2022 – Present

- Technical owner and architect of **ML Gateway**, the core orchestration layer for a high-throughput ML serving platform handling millions of requests per day; own the roadmap, PRDs/HLDs, and full lifecycle from pilot to production.
- Lead a team of **4 principal-level engineers** (previously 2 engineers and up to 8 contractors) driving platform performance and reliability: set technical direction, sequence the roadmap against competing launches, and review designs across caching, serving, and infrastructure.
- Designed and implemented **multi-region** model output caching with cascading invalidation pipelines, eliminating 40% of GPU inference requests, reducing median latency by 70%, and delivering an estimated **>\$10M/year in infrastructure cost savings** while limiting p99 latency impact to 2ms.
- Built a **self-service performance testing and optimization platform** that lets ML teams benchmark, profile, and tune their own workloads, catching regressions before production and removing my team as a bottleneck for the broader engineering org.
- Defined **SLO/SLI contracts adopted as platform-wide standards**; built the team's operational backbone (on-call rotation, triage, documentation program) that cut recurring incidents and ticket volume.
- Drove sustained cost and performance efficiency across the serving stack: **AWS Graviton migration**, Java 21 virtual threads, GC tuning, and authentication caching (5ms → 0.5ms), cutting end-to-end latency by >40%.
- Led coordination across **4+ teams** to bring a new LLM to production: selected TensorRT-LLM over vLLM, implemented prefix caching for KV-cache reuse, and migrated inter-service protocols to gRPC.
- Architected a distributed LLM-powered Query Understanding system serving **7M+ queries daily** at <55ms latency across streaming inference, batch pipelines (Spark/EMR), and online feature stores (Redis).

Machine Learning Engineer, Amazon May 2020 – Aug 2022

- Led **3 teams** through the architecture and launch of bid ceiling recommendations, an EMR/Glue service orchestrated via Airflow, improving bidder adoption by 5–10% for revenue-critical recommendations.
- Designed and optimized an ECS Java service for hourly budget allocation, cutting execution time **87.5%** (40ms → 5ms) through repeated profiling.

Quantitative Analyst, Central Bank of Canada Sep 2018 – Jan 2020

- Built quantitative models of market behavior identifying bond trading strategies saving **>\$300M/year**; published liquidity-premium research on Commonwealth MBS that informed policy.

PROJECTS

OctoMil — Distributed On-Device & Federated ML Platform, Founder / Engineer

- Built an end-to-end distributed platform solo: cross-platform inference SDKs (iOS, Android, browser/WebGPU, server), an orchestration server for fleet management and model aggregation, a CLI, and 9 privacy-preserving aggregation strategies.

EDUCATION

M.S. in Computer Science, University of Texas at Austin

Austin, TX

B.Com. in Finance & Economics, University of Toronto

Toronto, ON

SKILLS

Platform & Infrastructure: Distributed Systems, Java, Kubernetes (EKS), AWS (ECS/EMR/S3), Multi-Region Architecture, Cost Optimization, gRPC, Spark, Redis, OpenTelemetry, Go, Python

Leadership: Technical Strategy & Roadmapping, Leading Principal-Level Engineers & Multi-Team Initiatives, SLO/SLI Definition, On-Call & Incident Response, Self-Service Platform & Developer Enablement, Hiring & Mentorship

ML/AI Systems: LLM Inference & Serving (TensorRT-LLM, vLLM, ONNX), Quantization, KV/Prefix Caching, Federated Learning